

NANYANG TECHNOLOGICAL UNIVERSITY

School of Social Sciences

HE3002 Macroeconomics III

LECTURE 8

The IS-MPR-PC Model III

HOW DO INFLATION EXPECTATIONS CHANGE?

1. Let's refer back to Figure 2. We have seen that after the public's inflation expectations rise, the result is a fall in output below its natural rate and an increase in inflation, though this increase is smaller than had been expected by the public.
2. What happens next? How does the public's expectation of inflation change at this point?
3. Friedman's 1968 paper **The Role of Monetary Policy** suggested that people gradually adapt their expectations based on past outcomes for inflation.
4. Consider now a simple model of this idea of **adaptive expectations** by assuming that, each period, the expected level of inflation is simply equal to the level that prevailed last period. Formally, this can be written as:

$$\pi_t^e = \pi_{t-1} \quad (1)$$

5. Under this formulation of expectations, the Phillips curve becomes:

$$\pi_t^e = \pi_{t-1} + \gamma(y_t - y_t^*) + \epsilon_\pi^t \quad (2)$$

6. By rearranging equation (2), we get:

$$\pi_t^e - \pi_{t-1} = \gamma(y_t - y_t^*) + \epsilon_\pi^t \quad (3)$$

7. In other words, there should be a **positive** relationship between the change in inflation and the output gap.
8. There are various methods for measuring output gaps but one quick and easy method is to use the unemployment rate as an indication of what the output might be.
 - (a) If unemployment is high, then output is likely to be below its natural rate so the output gap is negative.

- (b) In contrast, a low unemployment rate is an indicator that the output gap is likely to be positive.
- 9. So if the adaptive expectations formulation of the Phillips curve was correct, then we would expect to see a **negative** relationship in the data between the change in inflation and the unemployment rate.

[**Figure 3:** Changes in inflation and unemployment, 1955-2013]

- 10. Figure 3 shows a scatter plot of data points on the change in inflation (measured as the four quarter percentage change in the price level minus the percentage change in the price level over the preceding four quarters) and the unemployment rate.
- 11. In contrast to the basic Phillips curve, this adaptive-expectations-style Phillips curve shows a clear and strong **negative** relationship between the change in inflation and the unemployment rate.
- 12. These results suggest that the adaptive expectations approach appears to provide a reasonable model of how people formulate inflation expectations.
- 13. That said, people are unlikely to simply use mechanical formulae to arrive at their expectations and one can imagine conditions in which people's inflation expectations could radically depart from what had happened in the past e.g. the appointment of a new central bank governor with a different approach to inflation, the adoption of a new currency or other major events.
- 14. Let's examine for now, however, how the IS-MP-PC model behaves when people have adaptive inflationary expectations.

ADJUSTMENT OF INFLATION EXPECTATIONS

- 1. After inflation expectations moved up to $\bar{\pi}$, the outcome was that inflation moved from π_1 (which is the central bank's inflation target) to π_2 .
- 2. If people follow adaptive expectations then the next period, they will set $\pi^e = \pi_2$.

[**Figure 4:** Inflation expectations adjusting back downwards]

- 3. Figure 4 shows what happens after this. The PC curve moves back downwards and inflation moves down to a lower level, denoted on this graph by π_3 .

4. If the public has adaptive expectations, then inflation and output gradually converge back to the point where output is at its natural rate and inflation equals the central bank's target rate.
5. If the public has adaptive expectations, how could inflation expectations just jump upwards? Rather than a random unexplained increase in inflation expectations, the more likely explanation for the Phillips curve shifting upwards because is temporary supply shocks, i.e. ϵ_t^π is positive for a number of periods.
6. Under adaptive expectations, the public becomes used to higher inflation and so the Phillips curve will remain above its long-run position even after the temporary supply shock has been reversed.

A TEMPORARY AGGREGATE DEMAND SHOCK

1. Having looked at what happens under adaptive expectations when the Phillips curve shifts, let's consider what happens when we have a temporary shock to aggregate demand, so ϵ_t^y takes a different value from zero, which means a shift in the IS-MPR curve.
2. We will illustrate a case where there is a shift towards a positive value of ϵ_t^y for a couple of periods but then it shifts back to zero.

[**Figure 5:** A temporary aggregate demand shock ($\epsilon_t^y > 0$)]

3. Figure 5 shows the immediate impact of a positive aggregate demand shock. Output and inflation both go up with inflation reaching the point denoted as π_2 . If the public has adaptive expectations, then this results in an increase in inflation expectations the following period.
4. When the aggregate demand shock persists but inflation expectations move up to match the previous period's inflation rate, the inflation rate now rises again.

[**Figure 6:** Reversal of aggregate demand shock leads to recession with high inflation]

5. Figure 6 shows what happens if the aggregate demand shock then reverses itself in the next period. The IS-MPR curve shifts back to its original position but the Phillips curve remains elevated. The result is a nasty combination of high inflation and output below its natural rate:
 - (a) An increase in output and inflation as the shock hits.
 - (b) A further increase in inflation as inflation expectations adjust upwards, accompanied by a decline in output.
 - (c) A decline in output and inflation as the shock disappears.
 - (d) A further decline in inflation accompanied by an increase in output as inflationary expectations gradually return to the central bank's target.
6. This shows that when the public has adaptive inflation expectations, temporary positive aggregate demand shocks lead to counter-clockwise loops.
7. It turns out that much of the data on inflation and output correspond to these kinds of movements.

Figure 7 is borrowed from notes on Stanford economist Charles I. Jones's website. They show the data from US on inflation and an estimated output gap from 1960 to 1983. The figure shows a number of periods of clear counter-cyclical movements.

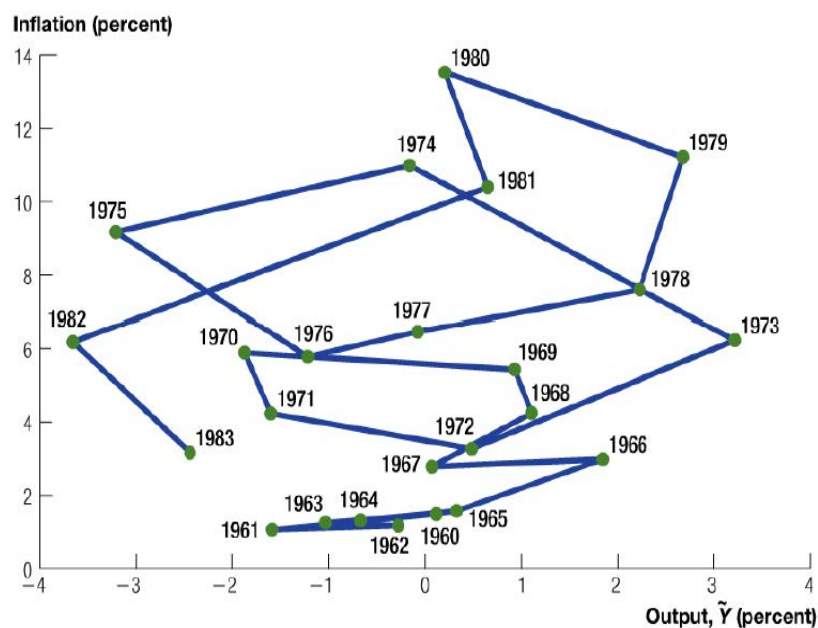


Figure 1: From Chad Jones: US inflation-output loops, 1960-1983

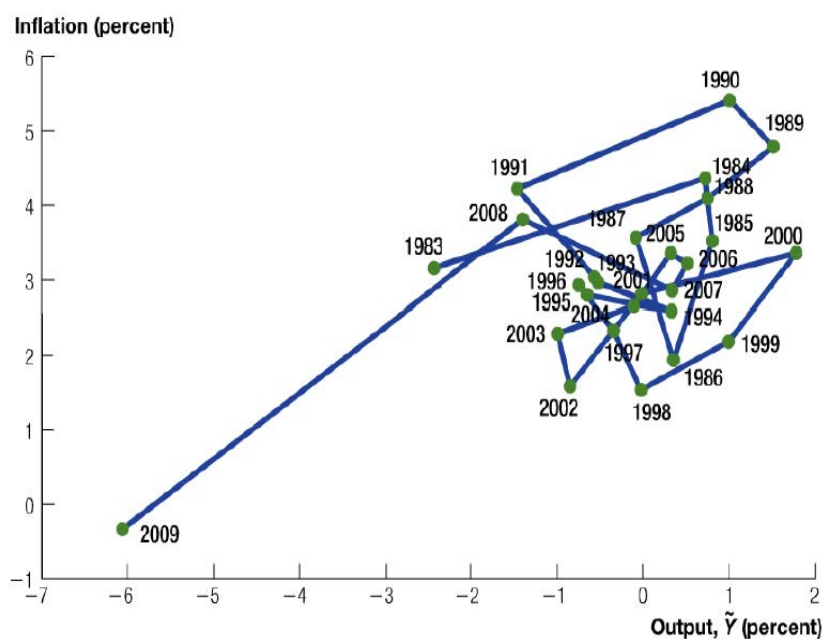


Figure 2: From Chad Jones: US inflation-output loops, 1983-2009

Figure 8 shows the same data from 1983-2009. This figure also shows some evidence counter-cyclical loops, though the movements are smaller than the for the pre-1983 period.

WHAT IF INFLATION EXPECTATIONS DON'T ADJUST?

1. The evidence suggests that adaptive expectations seems to be a reasonable model for how people have formulated their expectations of inflation.
2. And it can be argued that it is a fairly convincing model of how people behave: Most people don't have the time or knowledge to fully understand exactly what's going in the economy and anticipating that last year's conditions provide a guide to what will happen this year probably works well enough for most people.
3. Indeed, if the value of θ is relatively high, then inflation will only change slowly under adaptive expectations, making the adaptive expectations assumption more accurate.
4. All that said, it is also possible to imagine situations in which the public's inflation expectations are not formed adaptively.
5. For example, if the public believes that the central bank will always act to return inflation quickly towards its target, then they may assume that deviations from the target will be temporary.

[**Figure 9:** A temporary positive demand shock when inflation expectations don't change]

6. Figure 9 shows how the economy reacts to a temporary positive demand shock when inflation expectations don't change.
7. The outcome here is much nicer than the counter-clockwise cycle. There is no recession at any point, just a short period of output being above its natural rate and inflation being above its target, followed by a return to their starting levels.

IMPLICATIONS FOR MONETARY POLICY

1. The previous examples provide food for thought about what kind of monetary policy we would like a central bank to implement.
2. Do we want a **soft** central bank that limits the increase in real interest rates when inflation rises to protect the economy and which is not too concerned about getting inflation back to target quickly?
3. Or do we want a **tough** central bank that raises interest rates aggressively and is very concerned about getting inflation back to target?

4. Suppose we wish to avoid large recessions if possible. You might imagine the soft central bank would be more likely to deliver this.
5. However, our model says the exact opposite: Recessions are smaller and the economy less volatile when the central bank acts aggressively and is committed to returning inflation to its target rate.
6. We can see this in two results from the model:
 - (a) The economy responds better to shocks if the central bank reacts strongly to changes in inflation. We can see this from equations (4) and (5). Equations (3) and (4) together tell us that the more aggressively the central bank responds to inflation (the higher β_π is) the smaller is the increase in inflation in response to a rise in inflation expectations. Equation (5) then tells us this more aggressive approach means a smaller fall in output, i.e. a small recession. The process of a gradual return to the natural level of output and the central bank's target inflation rate will also be faster if the original changes were smaller. So recessions are smaller and shorter with an aggressive central bank.
 - (b) The more people believe that a central bank is maintaining its low inflation target, the less likely the economy is to go through boom-bust cycles. We can see this by comparing Figure 6 where inflation expectations shifts over time and Figure 9 which shows that happens when inflation expectations do not change after an expansionary shock.

IMPLICATIONS FOR CENTRAL BANK INSTITUTIONAL DESIGN

1. These results predict that we get better outcomes if we have a **tough** central bank that acts aggressively against inflation and which the public believes is committed to keeping the economy near its inflation target.
2. How can this outcome be achieved? The academic literature on this topic has suggested a number of different ways:
 - (a) **Political independence:** A central bank that plans for the long-term and does not worry about economic performance during election years is more likely to stick to a commitment to low inflation. So, independence from political control is an important way to reassure the public about the bank's credibility.
 - (b) **Conservative central bankers:** If the central banker is known to really dislike inflation and the public believes this, the economy gets closer to the ideal low inflation outcome even without commitment. So the government may choose to appoint a central banker who is more inflation-averse than they are.

- (c) **Consequence for bad inflation outcomes:** Introducing laws so that bad things happen to the central bankers when inflation is high is one way to make the public believe the they will commit to a low inflation rate.